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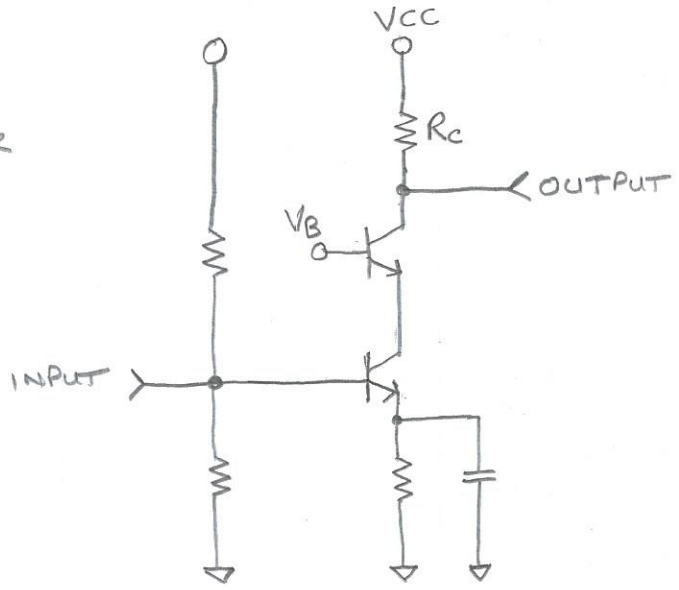
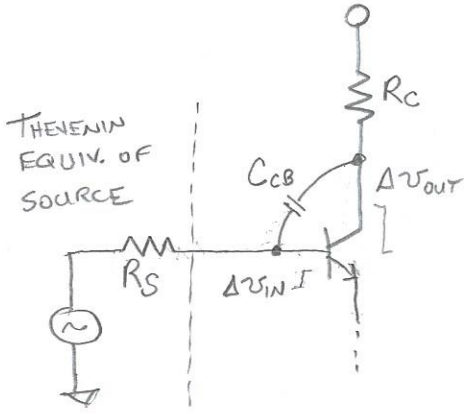
BASICS OF

THE CASCODE AMPLIFIER & THE MILLER EFFECT

- COMBINATION OF CE AMPLIFIER AND CB AMPLIFIER
- GAIN ≈ CE AMPLIFIER
- USED MAINLY TO REDUCE THE MILLER EFFECT



THE MILLER EFFECT



- MAIN BW LIMITING FACTOR
- C_{CB} IS MAGNIFIED BY GAIN
- BW LIMITED BY R·C FILTER ACTION OF R_S AND THE "MAGNIFIED" C_{CB}
- INPUT CAPACITANCE DUE TO C_{CB} LOOKS LIKE

$$C_{IN} = (|A_V| + 1) C_{CB}$$

$$g_m = \frac{I_C}{V_T} = \frac{1}{r_e}$$

$$A_V = -g_m R_C$$

CASCODE TO THE RESCUE

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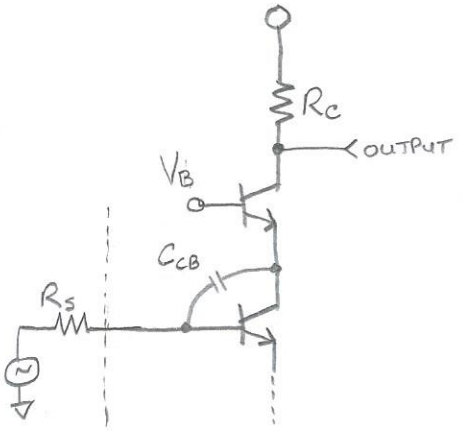


- CE STAGE GAIN IS $-g_m \cdot R_{IN-CB}$

$$R_{IN-CB} = r_e = \frac{1}{g_m}$$

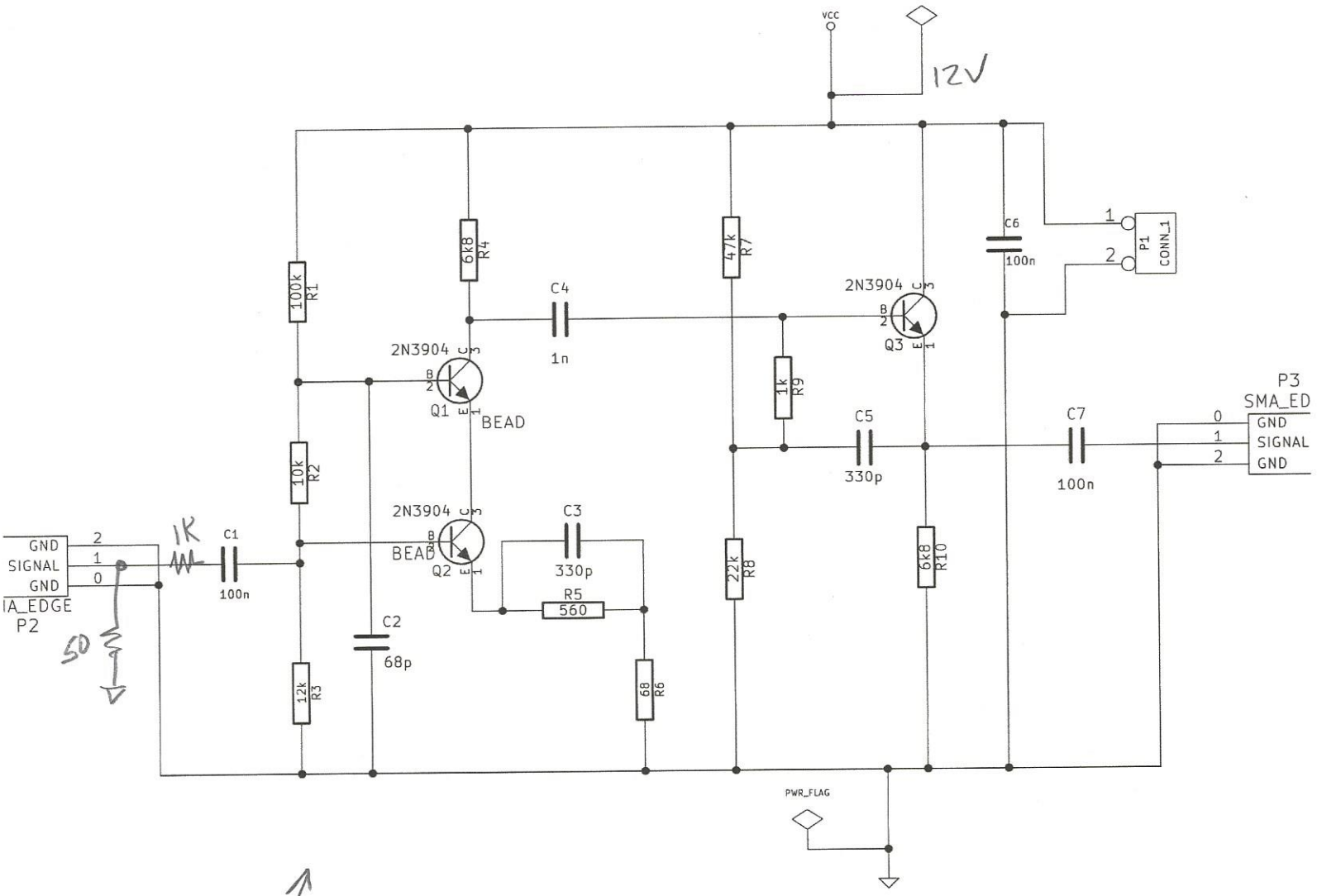
THUS; $A_{V_{CE}} = -g_m \cdot \frac{1}{g_m} = -1$

C_{CB} ONLY MAGNIFIED BY 2

- CB STAGE DOESN'T SUFFER FROM THE MILLER EFFECT

- $A_{V_{CB}} = g_m \cdot R_C$

THUS, OVERALL GAIN IS $\simeq -g_m \cdot R_C$



SCHEMATIC OF MARCUS' PCB (NOT MY DESIGN)

SEE MARCUS' BLOG FOR MORE INFO

(LINK IN VIDEO NOTES ON YOUTUBE)